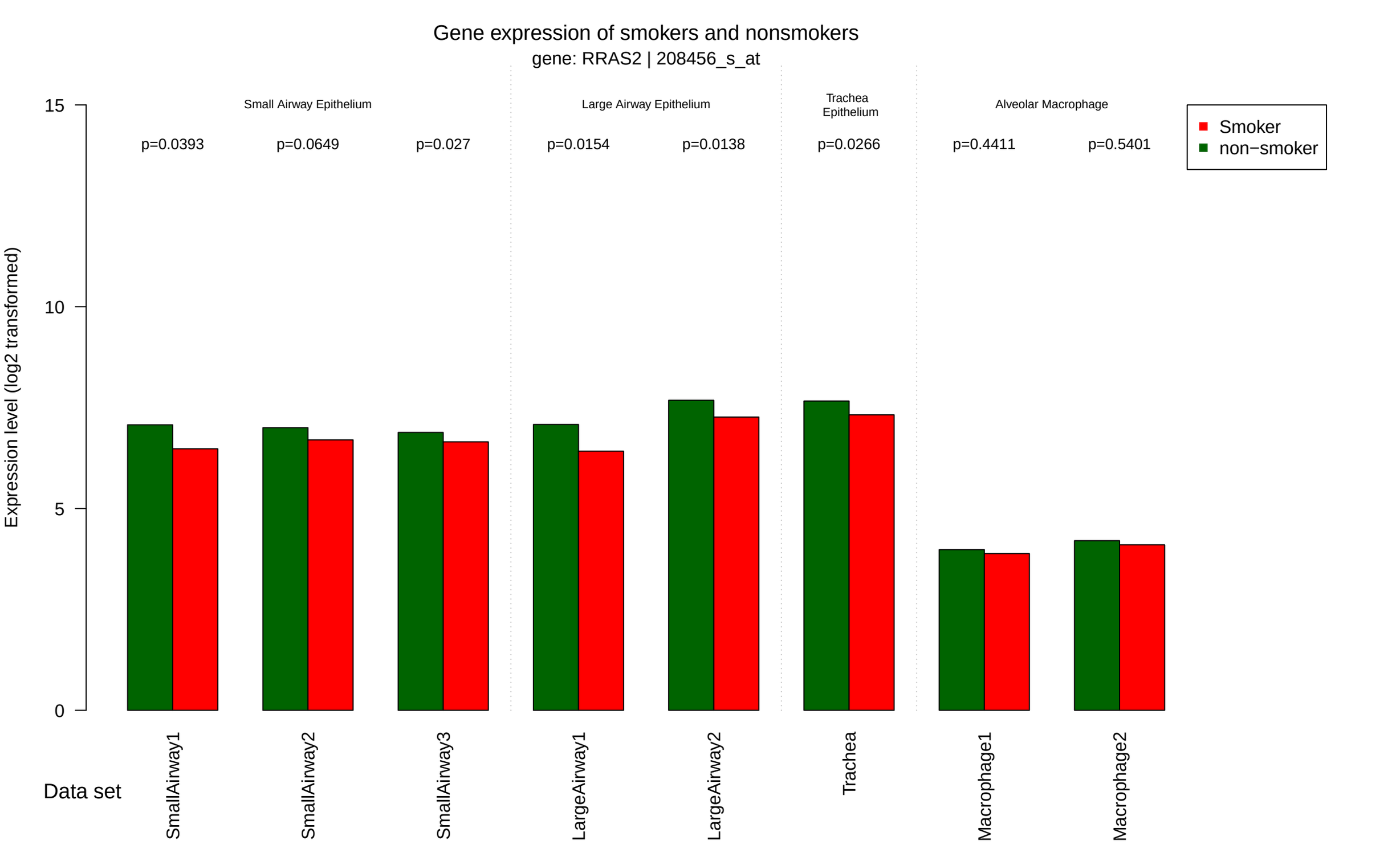
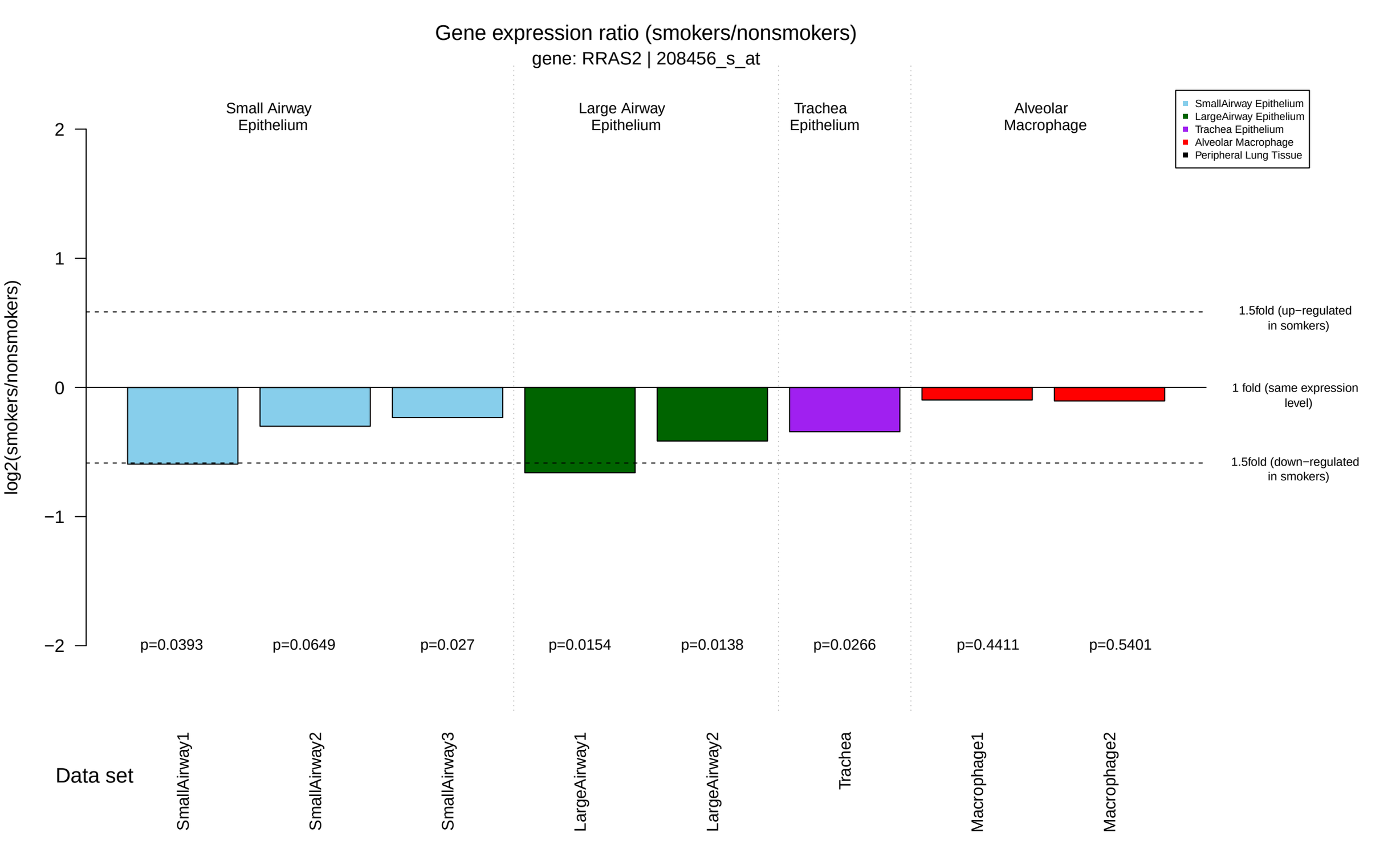
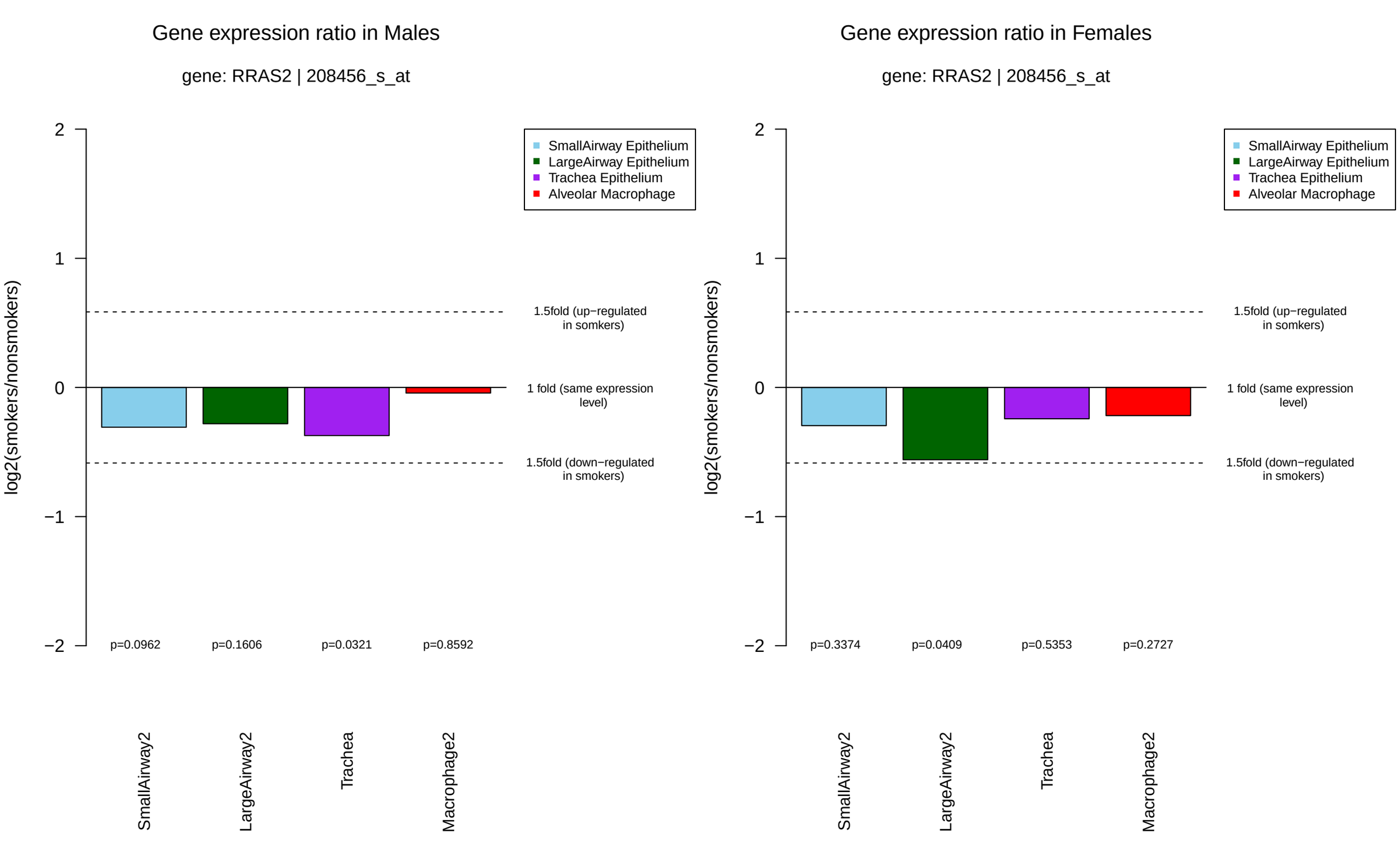
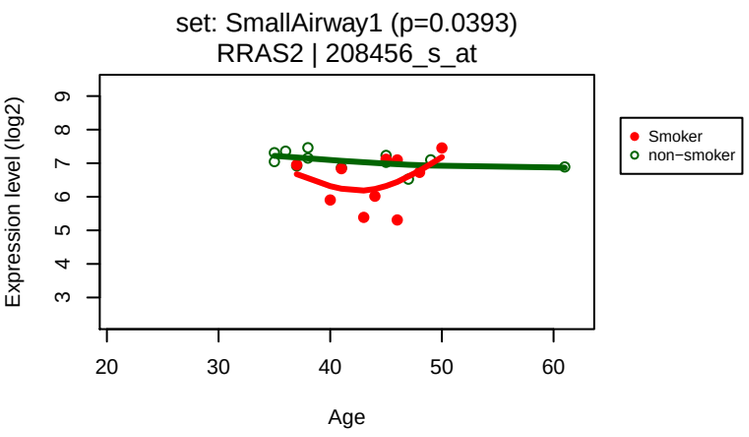
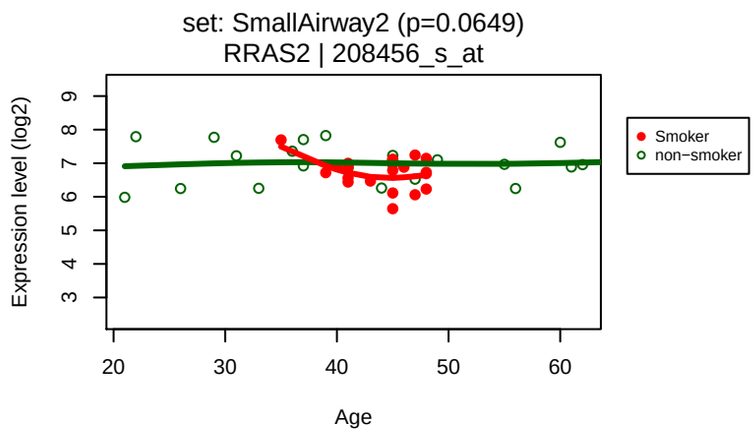
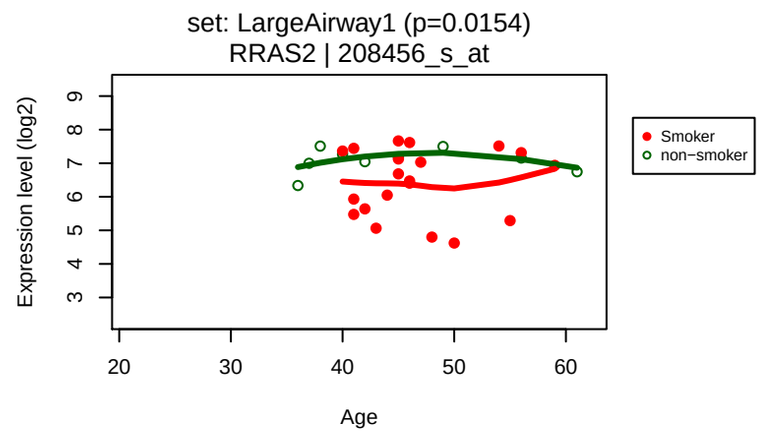

Gene expression of smokers and nonsmokers
gene: RRAS2 | 208456_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0393
p=0.0649
p=0.027
p=0.0154
p=0.0138
p=0.0266
p=0.4411
p=0.5401
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

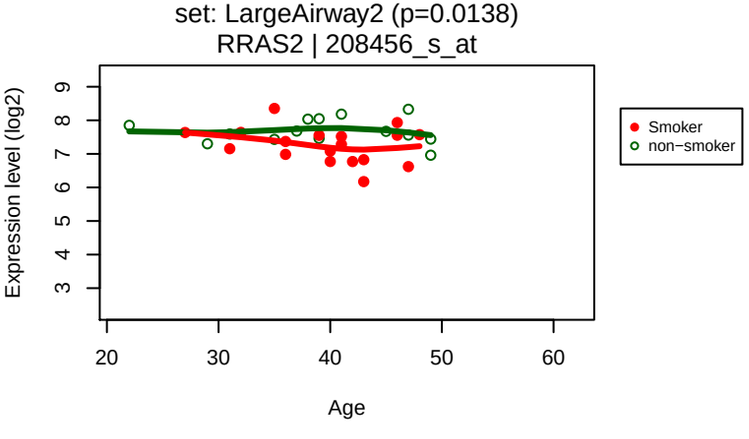

Gene expression ratio (smokers/nonsmokers)
gene: RRAS2 | 208456_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0393
p=0.0649
p=0.027
p=0.0154
p=0.0138
p=0.0266
p=0.4411
p=0.5401
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

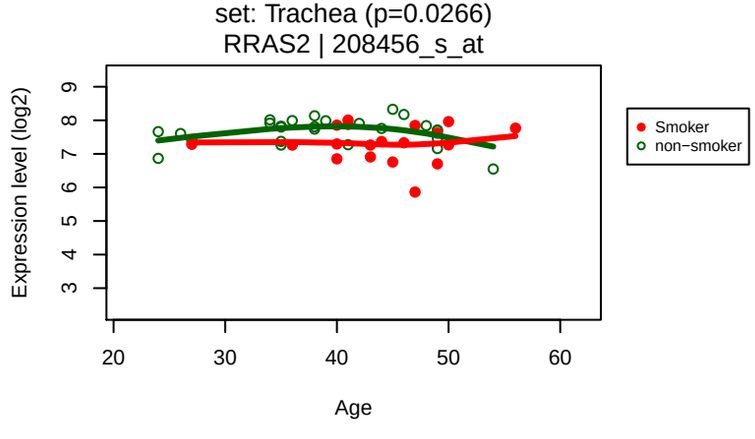

Gene expression ratio in Males
gene: RRAS2 | 208456_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0962
p=0.1606
p=0.0321
p=0.8592
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RRAS2 | 208456_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3374
p=0.0409
p=0.5353
p=0.2727
SmallAirway2
LargeAirway2
Trachea
Macrophage2

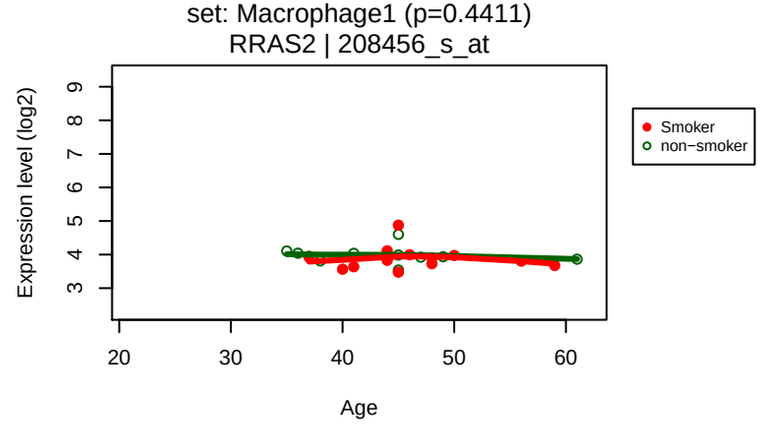

set: SmallAirway1 (p=0.0393)
RRAS2 | 208456_s_at
Expression level (log2)
Age
Smoker
non−smoker

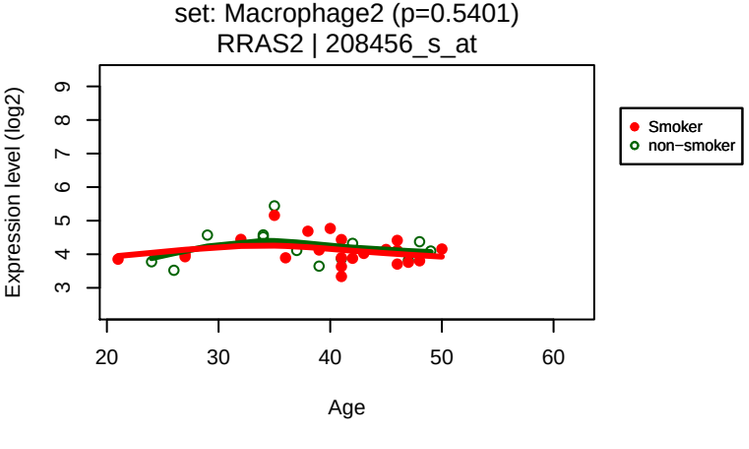

set: SmallAirway2 (p=0.0649)
RRAS2 | 208456_s_at
Expression level (log2)
Age
Smoker
non−smoker

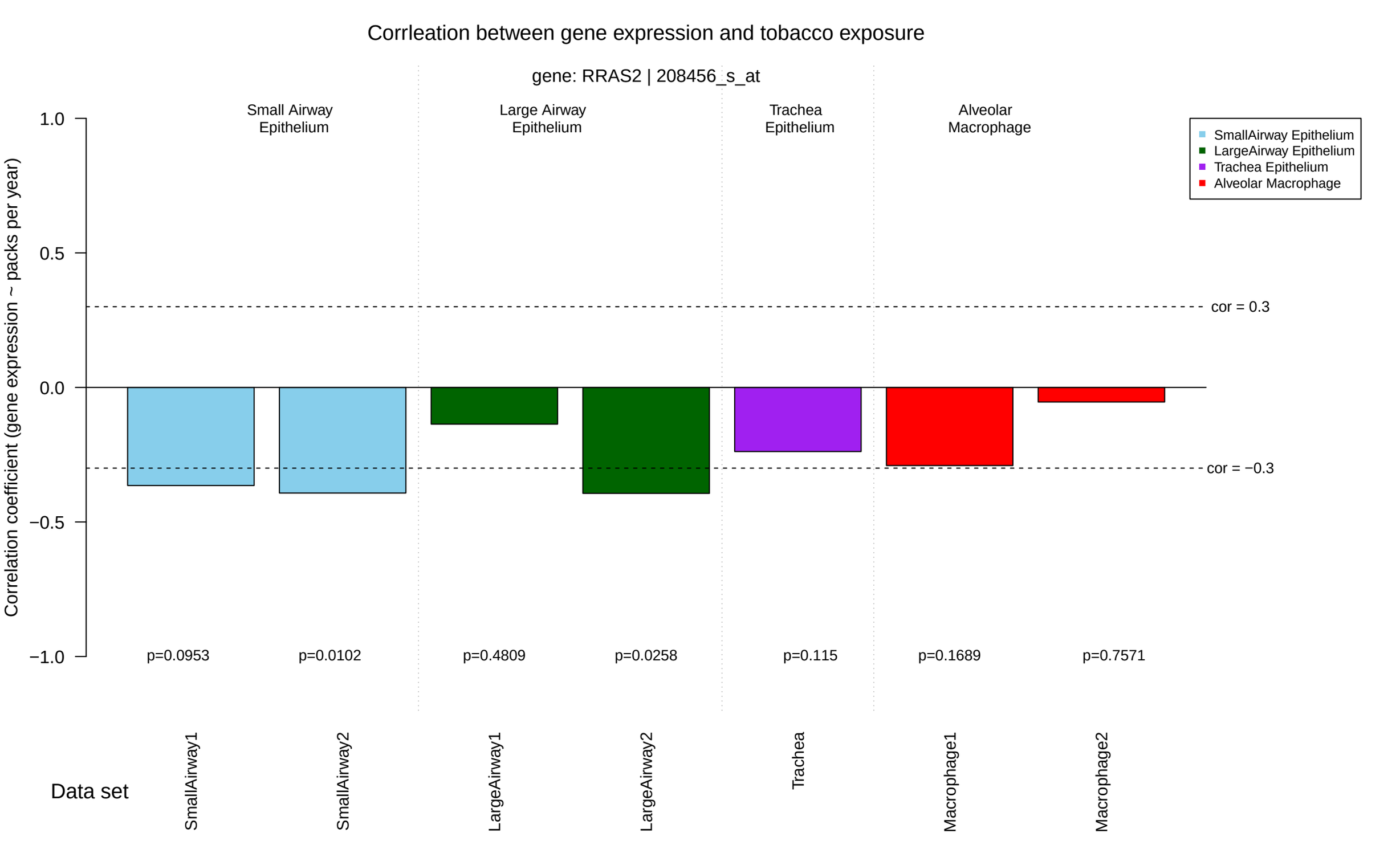

set: LargeAirway1 (p=0.0154)
RRAS2 | 208456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0138)
RRAS2 | 208456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0266)
RRAS2 | 208456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4411)
RRAS2 | 208456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5401)
RRAS2 | 208456_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RRAS2 | 208456_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0953
p=0.0102
p=0.4809
p=0.0258
p=0.115
p=0.1689
p=0.7571
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2